

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Seventh Semester of B. Tech (EC) Examination****November 2019****EC 444 Optical Communication****Date: 11.11.2019, Monday****Time: 01.30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer the question below [07]

- a A Double-heterojunction InGaAsP LED emitting at a peak wavelength of 1310nm has radiative and non radiative recombination time of 30 and 100ns, respectively. The drive current is 40mA. Find recombination lifetime, internal quantum efficiency and internal power level. **[03]**
- b Explain advantages of an optical fiber. **[04]**

Q – 2.a Explain Fabry-Perot Laser. [05]

Q – 2.b Explain Avalanche Photodiode. [05]

OR

- b Describe PIN Diode with necessary diagram. **[05]**
- c A given silicon avalanche photodiode has a quantum efficiency of 65% at wavelength of 900nm. Suppose 0.5μW of optical power produces a multiplied photocurrent of 10μA. Find multiplication M. **[04]**

Q – 3.a Which are the types of mechanical misalignment? [04]

- b Which are the lensing schemes for coupling improvement. **[06]**

OR

- b With time slot centers explain signal path through an optical link. **[06]**
- c With diagram explain which types of noise sources and disturbances occur at optical receiver? **[04]**

SECTION – II

- Q - 4.a Discuss situation of total internal reflection and derive snell's law with necessary diagram [03]
- b. Consider two silica fibers that are doped with 6 percent and 18 percent mode fractions of GeO_2 respectively. Compare the ultraviolet absorptions at wavelengths of 0.7 microm and 1.3 microm. [04]
- Q – 5. a Draw a typical WDM network containing various types of optical amplifiers and explain its operational principles. [05]
- b If attenuation coefficient is 0.4db/km, wavelength is 1310nm, cable length is 30km and input power is 200microW, then what is output power? [05]
- c Write a short note on multichannel transmission techniques. [04]

OR

- Q – 5. a A 2x2 biconical tapered fiber coupler has an input optical power level of $P_0=200\text{microW}$ at the other three ports are $P_1=90\text{microW}$, $P_2=85\text{microW}$ and $P_3=6.3\text{ nW}$. What are the coupling ratio, excess loss, insertion losses, and return loss for this coupler? [05]
- b If source power is -42db, received power -13db, connector loss 1db, attenuation coefficient 3.5db/km, system margin 6 db then find out possible transmission path in km. [05]
- c Write a short note on optical circulators. [04]
- Q – 6.a Describe signal dispersion in fibers. Explain intramodal dispersion in detail. [05]
- b Write a short note on outside vapor phase oxidation with process diagram. [05]
- c Explain basic optical fiber network topologies with figures. [04]

OR

- Q – 6.a Consider the following two multimode fibers: (i) a step-index fiber with a core index 1.458 and a core cladding index difference 0.01 (ii) a parabolic -profile graded-index fiber with the same values of indexes. Compare the rms pulse broadening per kilometer for these two fibers. [05]
- b Write a short note on plasma activated chemical vapor deposition with process diagram. [05]
- c Explain and compare basic formats of STS-N SONET frame and STM-N SDH frame. [04]
